Question 5: Initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-1-

1. Initialize, stage and commit

init, stage, commit

```
cd ~/ensp461-lab1
git init -b main
git add README.md .gitignore app/app.py app/requirements.txt app/tests/test_app.py
git commit -m "Initial commit: sample Flask application with unit tests"
git add deploy/deploy.sh
git commit -m "Add automated deployment script with health verification"
```

2. Branch, feature work and merge

branch and merge (--no-ff)

```
git checkout -b feature/app-info-endpoint
# implement /info endpoint + unit test on the branch
git add app/ && git commit -m "Add /info endpoint with application metadata and unit test"
git checkout main
git merge --no-ff feature/app-info-endpoint -m "Merge feature/app-info-endpoint: add /info \
      endpoint to main"
```

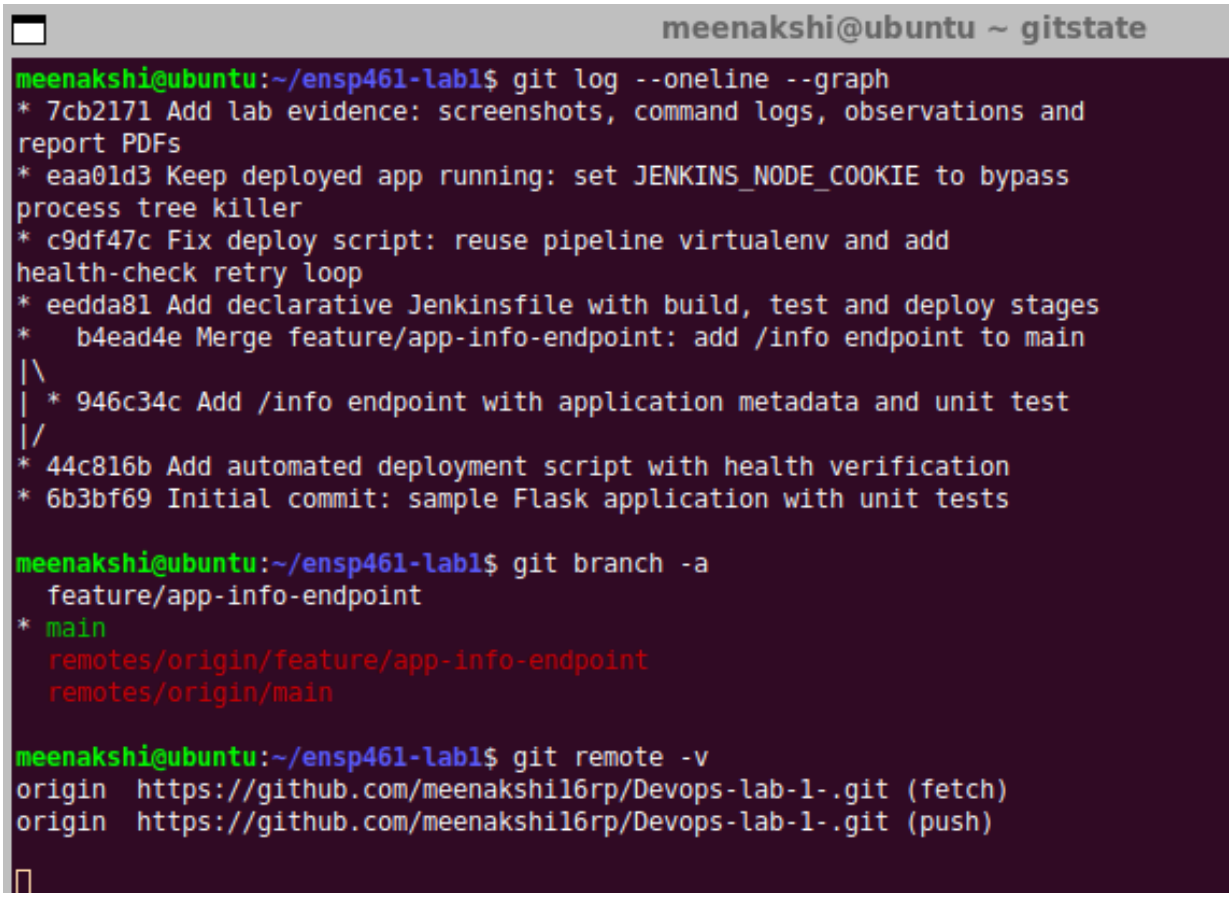


Figure: Ubuntu terminal: commit history after the merge (git log --graph)

3. Push to GitHub

remote and push

```
git remote add origin https://github.com/meenakshi16rp/Devops-lab-1-.git
git push -u origin main
git push -u origin feature/app-info-endpoint
```

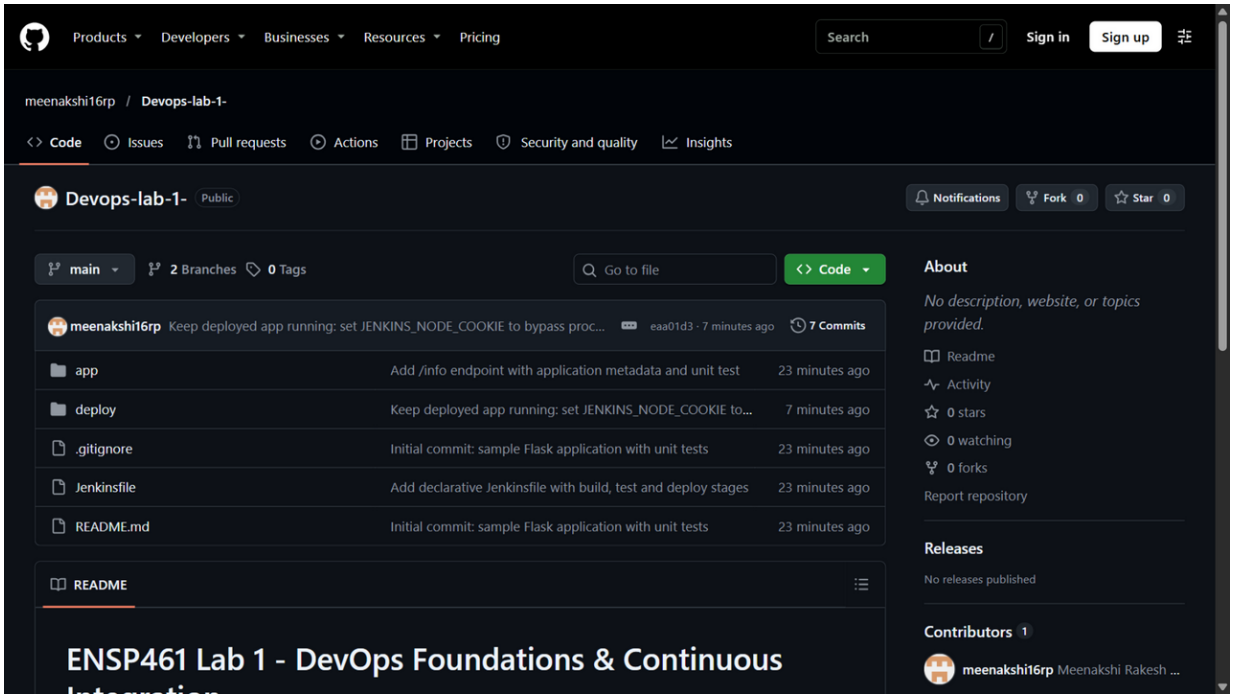


Figure: Repository on GitHub: 7 commits, 2 branches, Jenkinsfile and README visible

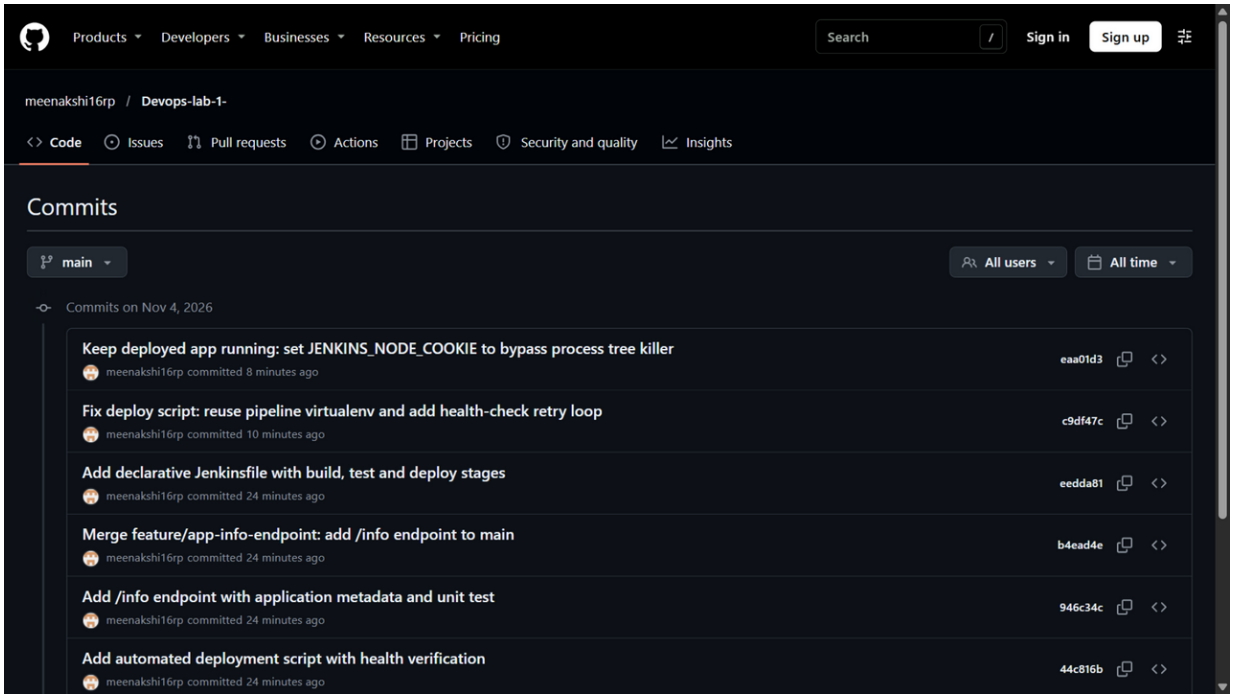


Figure: Commit history on GitHub - merge commit and feature branch visible

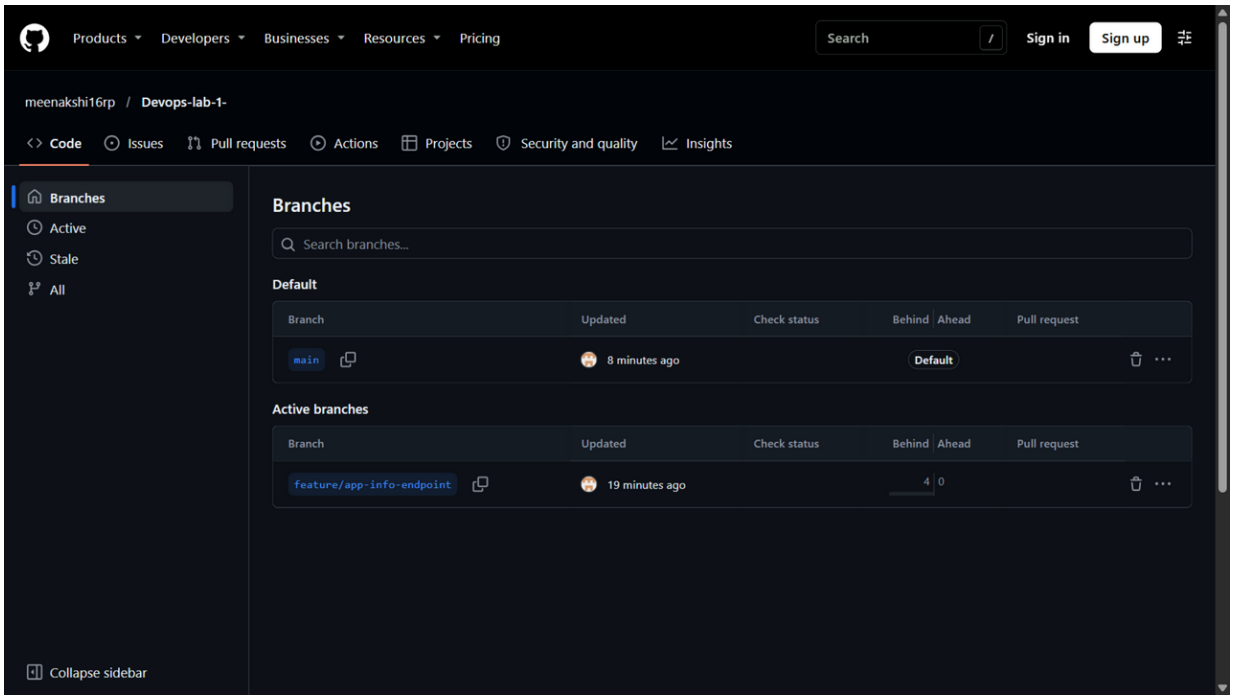


Figure: Branches page: main (default) and feature/app-info-endpoint

4. Observations

- --no-ff merging keeps the feature branch visible in history, which documents the workflow.
- Every commit message describes what changed so the log reads as a narrative.
- The feature branch remains on GitHub as evidence of the branching workflow.
- credential.helper store keeps the PAT outside the repository (in ~/.git-credentials, chmod 600).